



INFLATION FORECASTING IN INDIA USING TIME SERIES AND MACHINE LEARNING

**PROJECT REPORT SUBMITTED TO
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MASTER OF SCIENCE DEGREE IN STATISTICS**

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CERTIFICATE

This is to certify that the report entitled **INFLATION FORECASTING IN INDIA USING TIME SERIES AND MACHINE LEARNING** submitted by **AJITH KC**, to Mahatma Gandhi University, Kottayam , in partial fulfillment of the requirements for the award of the Degree in Master of Science in Statistics is a bonafide record of the project work carried out by his under my guidance and supervision. This report in any form has not been submitted to any other University or Institute for the award of any degree or diploma.

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DECLARATION

I, **AJITH KC**, a final-year M.Sc. Statistics student of Nirmala College, Muvattupuzha (Autonomous), Affiliated by Mahatma Gandhi University Kottayam, Kerala, hereby declare that the project report entitled **INFLATION FORECASTING IN INDIA USING TIME SERIES AND MACHINE LEARNING**, submitted for partial fulfillment of the requirements for the award of the degree of Master of Science under Mahatma Gandhi University, Kottayam, Kerala is a bonafide work done by me. This submission represents my ideas in my own words and where the words of others have been included, I have adequately and accurately cited and referenced the original sources. I also declare that I have adhered to the ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in my submission. I understand that any violation of the above will be cause for disciplinary action by the institute and/or the University can also evoke penal action from the sources that have thus not been properly cited or from whom proper permission has not been obtained. This report has not previously formed the basis for the award of any degree, diploma, or similar title of any other University.

Place : Muvattupuzha.

AJITH KC

Date : 17/04/2026

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ABSTRACT

Inflation management constitutes a fundamental pillar of India's monetary policy, with far-reaching implications for economic stability and sustainable growth. This study aims to forecast Consumer Price Index (CPI) inflation for the year 2026, with particular emphasis on the transition to the Flexible Inflation Targeting (FIT) framework. A key feature of the analysis is the identification of a structural break in September 2016, marking the onset of a more stable and policy-driven inflation regime. By concentrating on this period, the study seeks to generate a realistic and policy-consistent inflation outlook aligned with the Reserve Bank of India's objectives.

The study employs monthly data spanning 2012 to 2025, with CPI considered as the target variable and key macroeconomic indicators—namely Wholesale Price Index (WPI) inflation and international crude oil prices—incorporated as exogenous variables. A comparative evaluation is conducted between classical time series models and advanced machine learning techniques. Specifically, an ARIMAX(2,0,2) model and a Support Vector Regression (SVR) model with a radial basis function (RBF) kernel are implemented. The findings reveal that the machine learning approach demonstrates superior predictive performance, whereas the ARIMAX model offers greater interpretability and economic relevance.

The results indicate that inflation in India is transitioning toward a phase of relative stability following short-term volatility observed in late 2025. The forecast for 2026 suggests that inflation is likely to remain within the range of 2.1% to 3.1%, consistent with the prescribed target band. The significance of autoregressive components underscores the persistence and anchoring of inflation expectations under the current policy framework. Overall, the study highlights that incorporating structural breaks alongside a hybrid modeling framework enhances forecasting accuracy and provides meaningful insights for policy formulation and economic decision-making.

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CHAPTER 01

INTRODUCTION

1.1 BACKGROUND AND MOTIVATION

Inflation is a key macroeconomic indicator that reflects the overall price stability within an economy. In a developing country like India, maintaining stable inflation is essential for sustainable economic growth, effective resource allocation, and the preservation of purchasing power. High and volatile inflation can adversely affect investment decisions, income distribution, and overall economic confidence, making accurate inflation forecasting a critical task for policymakers.

In recent years, India has adopted the Flexible Inflation Targeting (FIT) framework, which places primary emphasis on controlling inflation within a specified target band. This policy shift has enhanced the importance of reliable and timely inflation forecasts for effective monetary policy formulation, particularly for institutions such as the Reserve Bank of India (RBI). However, inflation dynamics in India are complex and influenced by multiple factors, including domestic price pressures, global commodity prices, and exchange rate fluctuations.

Traditional time series models, such as ARIMA and SARIMA, have been widely used for inflation forecasting due to their ability to capture temporal patterns and seasonality. However, these models may have limitations in capturing non-linear relationships and external shocks. In contrast, modern machine learning techniques offer improved predictive performance by modeling complex and non-linear interactions among variables.

Motivated by these considerations, this study aims to develop a comprehensive inflation forecasting framework by combining classical econometric approaches with machine learning techniques. By incorporating key macroeconomic variables such as Wholesale Price Index (WPI) and crude oil prices, the study seeks to better understand the drivers of inflation and enhance forecasting accuracy. The comparative analysis of these approaches provides valuable insights into the trade-off between interpretability and predictive performance, thereby contributing to more informed economic decision-making.

1.2 OBJECTIVES OF THE STUDY

- To analyze the historical trends and patterns of CPI inflation in India from 2012 to 2025 to understand the underlying volatility.
- To investigate potential regime shifts in the inflation series and statistically determine the most appropriate time-period for high-precision forecasting.
- To evaluate a diverse range of forecasting models, including classical statistical methods and multiple Machine Learning algorithms, to identify the most robust predictive architecture.
- To quantify the impact of exogenous drivers, specifically International Crude Oil prices and the Wholesale Price Index (WPI), on the selected champion models.
- To perform comparative model diagnostics using RMSE, MAE, and R^2 to justify the selection of the final forecasting framework.
- To generate a reliable 2026 inflation forecast based on the best-performing model and assess its implications within the RBI's mandated tolerance band.

1.3 SOFTWARE TOOLS AND PROGRAMMING PACKAGES

The implementation of this study relied on a combination of data engineering tools and statistical programming environments to ensure data integrity and model accuracy.

1.3.1 Data Processing Tools

- **Microsoft Excel (Power Query):**
Microsoft Excel was utilized for the initial extraction and preprocessing of datasets sourced from the Reserve Bank of India (RBI) and the Ministry of Statistics and Programme Implementation (MOSPI). Power Query played a crucial role in integrating multiple time-series files, addressing missing values, and transforming the raw data into a structured and analysis-ready format.
- **CSV (Comma-Separated Values):**
CSV files were used as the primary format for transferring data between Excel and Python, ensuring compatibility and ease of use.

1.3.2 Statistical Programming

The main analysis was carried out using Python, which provides a wide range of tools for time-series analysis and machine learning. The key libraries used in this study are as follows:

- **Pandas and NumPy:**
Used for data handling, time-series indexing, and numerical computations, including the calculation of inflation rates.
- **Statsmodels:**
Applied for econometric analysis, including ARIMA and ARIMAX(2,0,2) models, as well as statistical tests such as the Augmented Dickey-Fuller (ADF) test and Ljung–Box test.
- **Scikit-learn:**
Used for implementing machine learning models, particularly Support Vector Regression (SVR), along with other models such as Random Forest and Gradient Boosting. It was also used for data scaling and model evaluation.
- **Matplotlib and Seaborn:**
Used to create clear and informative visualizations, including time-series plots, residual analysis, and forecast comparisons.
- **SciPy:**
Used for statistical testing, including the Shapiro–Wilk test for normality and checks for heteroscedasticity.

1.4 SCOPE AND SIGNIFICANCE

The scope of this study focuses on the monthly Consumer Price Index (CPI – Combined) of India from 2012 to 2025, with particular emphasis on the post-2016 period to capture the shift toward the inflation targeting framework. The analysis considers the influence of key macroeconomic variables such as the Wholesale Price Index (WPI) and international crude oil prices, using ARIMAX and Support Vector Regression (SVR) models.

The significance of the study lies in the following key aspects:

- Highlights the importance of structural changes in inflation analysis
- Identifies the September 2016 structural break for improved forecasting
- Provides a practical inflation outlook for 2026
- Offers useful insights for policy and economic planning

CHAPTER 02

EXPLORATORY DATA ANALYSIS (EDA)

2.1 INTRODUCTION TO EDA

Exploratory Data Analysis (EDA) is a fundamental step in any data analysis or modeling process. It involves systematically examining the dataset to understand its structure, identify patterns, detect anomalies, and assess relationships between variables before applying statistical or machine learning models.

In this study, EDA was conducted on the cleaned macroeconomic dataset, including CPI inflation, WPI inflation, crude oil prices, and exchange rate data. The purpose was to gain meaningful insights into the behavior of these variables and to ensure data quality prior to model development.

The key objectives of EDA in this study are as follows:

- To understand the distribution and range of each variable
- To identify missing values or inconsistencies in the data
- To detect outliers and assess their impact
- To examine relationships between the target variable (CPI) and explanatory variables
- To identify trends, seasonality, and structural changes in the data

Through this analysis, important insights were obtained regarding inflation dynamics and the influence of domestic and global factors. These findings provided a strong foundation for selecting appropriate modeling techniques and improving forecast accuracy.

2.2 DATA DESCRIPTION AND SOURCES

This chapter provides a detailed overview of the datasets utilized in the study, their origins, and the specific variables selected to model India's inflation trajectory.

2.2.1 Data Sources

The secondary data for this research was meticulously collected from official government and international financial databases to ensure maximum reliability and transparency.

- Consumer Price Index (CPI-Combined): Sourced from the Ministry of Statistics and Programme Implementation (MOSPI) via the Database on Indian Economy (DBIE) portal.
- Wholesale Price Index (WPI): Extracted from the Office of the Economic Adviser, Government of India, representing the primary measure of producer-level inflation.
- International Crude Oil Prices: Monthly Brent Crude oil prices (in USD per barrel) were obtained from the Federal Reserve Economic Data (FRED) and Investing.com, representing the critical global supply-side driver for the Indian economy.
- Exchange Rate (INR/USD): Monthly average exchange rates were sourced from the Reserve Bank of India (RBI), representing the external value of the Rupee

2.2.2 Variable Description

The study employs a multivariate approach, selecting three primary indicators spanning the period from January 2012 to December 2025:

1. CPI (Target Variable): The Consumer Price Index (Combined) with a base year of 2012. It serves as the primary gauge for retail inflation and the official target for the RBI's monetary policy.
2. WPI (Exogenous): Included to capture the "pass-through" effect from the manufacturing and wholesale sectors to retail prices.
3. Crude Oil Price (Exogenous): Included due to India's high import dependency, where fluctuations in global energy costs act as a precursor to domestic transportation and production cost increases.
4. Exchange Rate (Exogenous): Included to account for "Imported Inflation." As the Rupee depreciates against the Dollar, the landing cost of essential imports (like oil and electronics) increases, directly affecting the domestic CPI.

2.2.3 Summary Table of Data Sources

Variable	Frequency	Source	Significance
CPI (Combined)	Monthly	MOSPI	Primary Target (Inflation Gauge)
WPI	Monthly	MOSPI	Leading Indicator (Supply Side)
Brent Crude Oil	Monthly	Investing.com	External Global Shock Proxy
Exchange Rate	Monthly	Reserve Bank of India	Imported Inflation & Currency Risk

Table 2.1

2.3 DATA HANDLING AND PRE-PROCESSING

The raw data collected from multiple sources required systematic processing to ensure consistency, reliability, and suitability for statistical modeling. The data handling process was carried out in three main stages:

2.3.1 Data Extraction and Cleaning

Initial data preparation was performed using Microsoft Excel, particularly the Power Query tool, to address structural differences across datasets obtained from RBI, MOSPI, and FRED.

- **Merging and Alignment:**
The key variables—CPI, WPI, crude oil prices, and exchange rate—were merged into a unified dataset using a common monthly time index.
- **Handling Missing Values:**
Missing observations and gaps in reporting were identified and treated to ensure a continuous time series from January 2012 to December 2025.
- **Format Standardization:**
Data formats were standardized by ensuring consistent numerical

representation and converting date variables into a machine-readable format (ISO 8601).

2.4 SUMMARY STATISTICS

Statistic	CPI(%)	WPI(%)	Crude oil price	Exchange rate
Mean	5.31	3.08	67.31	71.80
SD	2.11	4.70	20.13	8.80
Minimum	0.25	-6.14	18.84	53.77
Q_1	3.81	0.23	51.25	64.48
Median	5.09	2.51	65.31	71.19
Q_2	6.55	4.81	80.55	79.92
Maximum	11.51	16.63	114.67	88.83

Table 2.2

The summary statistics provide important insights into the behavior of the variables over the study period. CPI inflation averages around 5.31% and shows moderate variability, indicating relative stability. In contrast, WPI inflation is more volatile, with a wider range and occasional negative values.

Crude oil prices exhibit the highest variability, reflecting strong exposure to global market fluctuations, while the exchange rate shows a steady upward trend, indicating gradual currency depreciation.

Overall, the wide range across variables highlights the presence of both stable periods and economic shocks, supporting the need for robust modeling techniques.

2.5 VISUAL EXPLORATIONS

Visualization plays a crucial role in Exploratory Data Analysis (EDA) by facilitating the interpretation of data distributions, identification of patterns, and detection of relationships among variables. In this study, various graphical techniques were employed using Python libraries such as Matplotlib and Seaborn to gain a deeper understanding of the dataset. These visualizations help in revealing trends, volatility, and interdependencies among key variables, thereby supporting more informed modeling decisions.

2.5.1 Time Series Plots

Time series plots are used to visualize how a variable changes over time, helping to identify trends, patterns, and fluctuations. They are particularly useful for detecting long-term movements, seasonality, and sudden shocks in the data.

- **CPI and WPI Inflation Over Time**

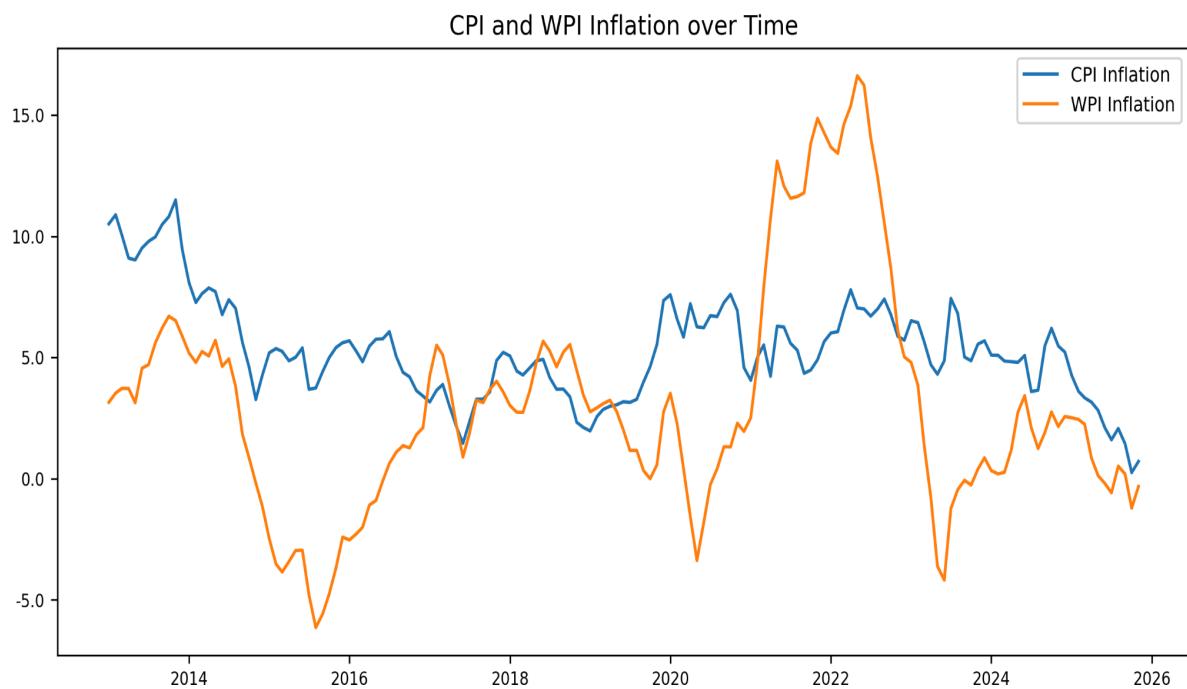


Figure 2.1

The time-series plot of CPI and WPI inflation reveals important dynamics in price movements. CPI inflation shows a declining trend from the high levels observed during 2013–2014, followed by relative stability after 2016. In contrast, WPI inflation exhibits higher volatility, including negative values around 2015–2016.

A noticeable pattern is that changes in WPI often precede movements in CPI, indicating a lagged transmission from wholesale to retail prices. This supports the inclusion of WPI as a leading indicator in the forecasting models.

- **Crude Oil Price Trend**

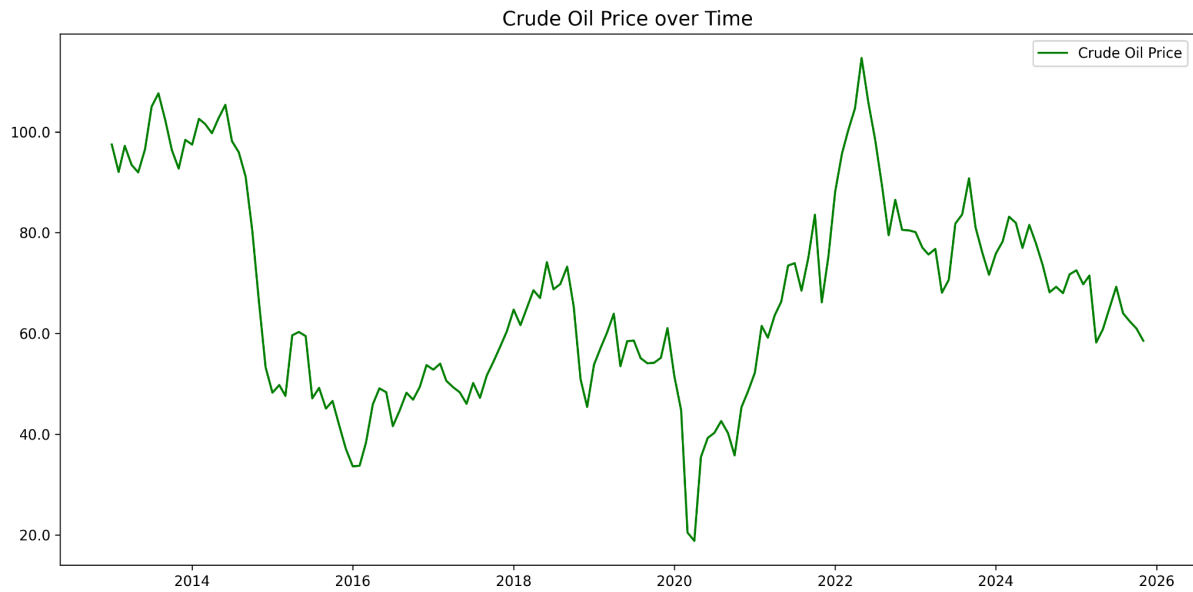
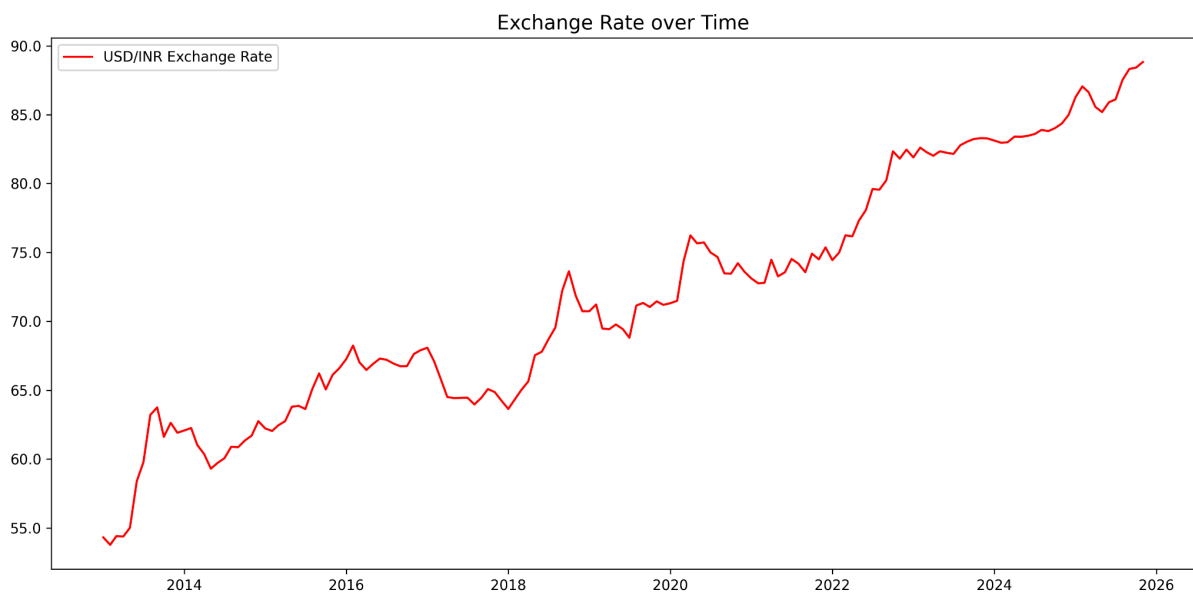


Figure 2.2

The crude oil price series shows significant fluctuations over the study period. A sharp decline is observed around 2020, followed by a strong recovery and peak during 2021–2022. These fluctuations correspond to global economic events and have a direct impact on inflation through transportation and production costs. The observed volatility highlights the role of crude oil as a key external driver of inflation in India.

- **Exchange Rate Trend**



The exchange rate trend indicates a gradual depreciation of the Indian Rupee over time. This weakening of the domestic currency increases the cost of imports, contributing to inflationary pressures.

The steady upward movement in the exchange rate suggests a persistent external influence on inflation, particularly during periods of global economic uncertainty.

2.5.2 Histograms

Histograms are used to visualize the distribution and frequency of data values within a dataset. They help in identifying patterns such as skewness, dispersion, and the presence of outliers in the variables.

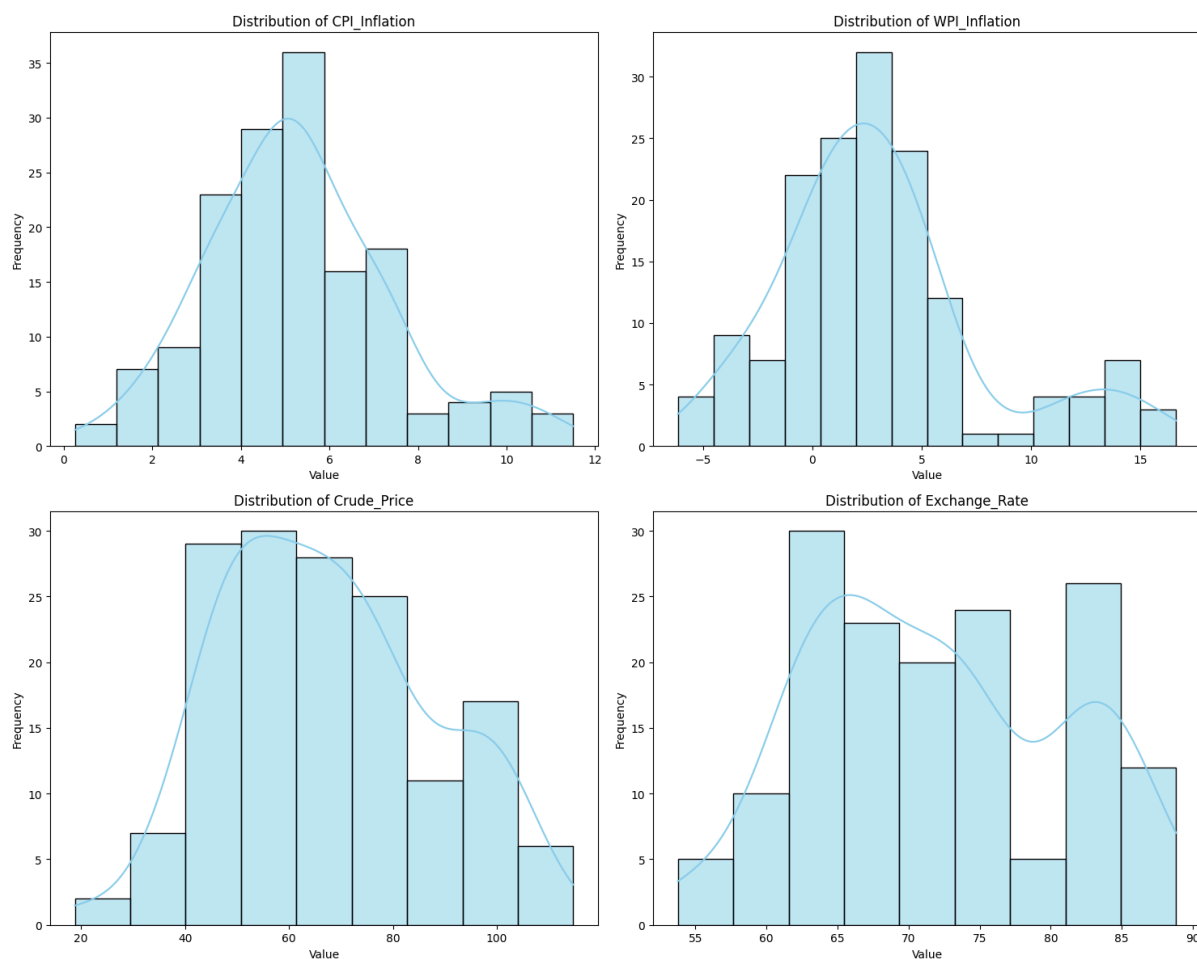


Figure 2.4

- The distribution of CPI inflation appears approximately normal with slight right skewness, indicating that most values are concentrated around the mid-range (approximately 3%–7%). A few higher values extend the right tail, representing periods of elevated inflation. This suggests that CPI inflation remains relatively stable over time, with occasional spikes due to economic shocks.

- The distribution of WPI inflation shows greater dispersion and noticeable skewness, with values extending into both negative and high positive ranges. The presence of negative values indicates periods of deflation at the wholesale level. Compared to CPI, WPI is clearly more volatile, reflecting stronger fluctuations in producer prices.
- The crude oil price distribution is wide and moderately skewed, indicating significant variability over the sample period. Most observations are clustered in the mid-range (around 50–80 USD), while extreme values reflect sharp global price movements. This highlights the high volatility of oil prices, driven by global economic and geopolitical factor
- The exchange rate distribution shows a moderate spread with slight right skewness, with most values concentrated between 60 and 80 INR/USD. The upper tail reflects periods of higher depreciation. This indicates a gradual upward trend (depreciation) with moderate fluctuations over time.

2.5.3 Box Plots

Boxplots were generated for each feature to identify outliers and understand the spread of data

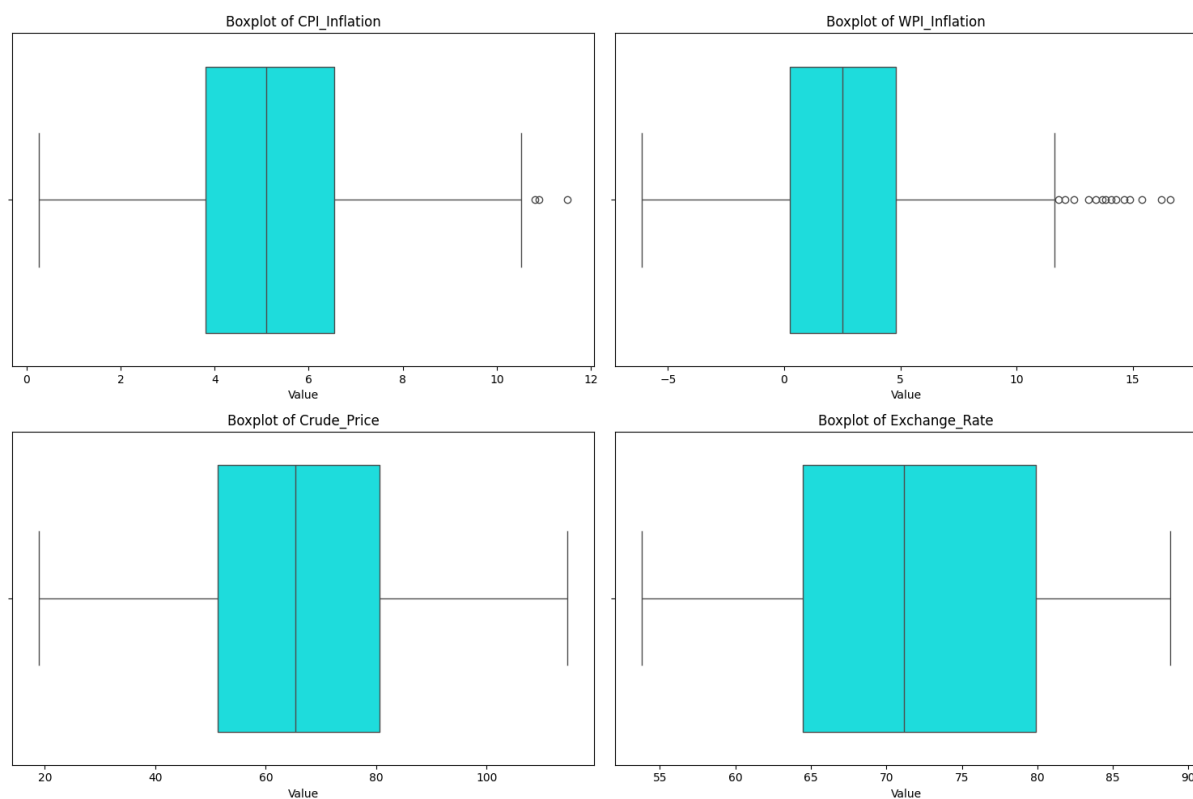


Figure 2.5

- The median CPI is around 5%, with a relatively narrow interquartile range, indicating stable inflation. A few upper outliers reflect periods of high inflation, and the slight positive skew suggests occasional upward shocks.
- WPI has a lower median but much wider dispersion, showing higher volatility. The presence of both positive and negative outliers indicates periods of sharp price increases as well as deflation.
- Oil prices exhibit the highest variability, with a wide range and large spread. The absence of outliers suggests that even extreme values are part of normal fluctuations driven by global market conditions.
- The exchange rate shows a wide range with no significant outliers, indicating a gradual and consistent depreciation of the currency over time rather than sudden shocks.

2.5.4 Correlation Heatmap

A correlation heatmap was generated to examine linear relationships among the variables. Important relationships observed

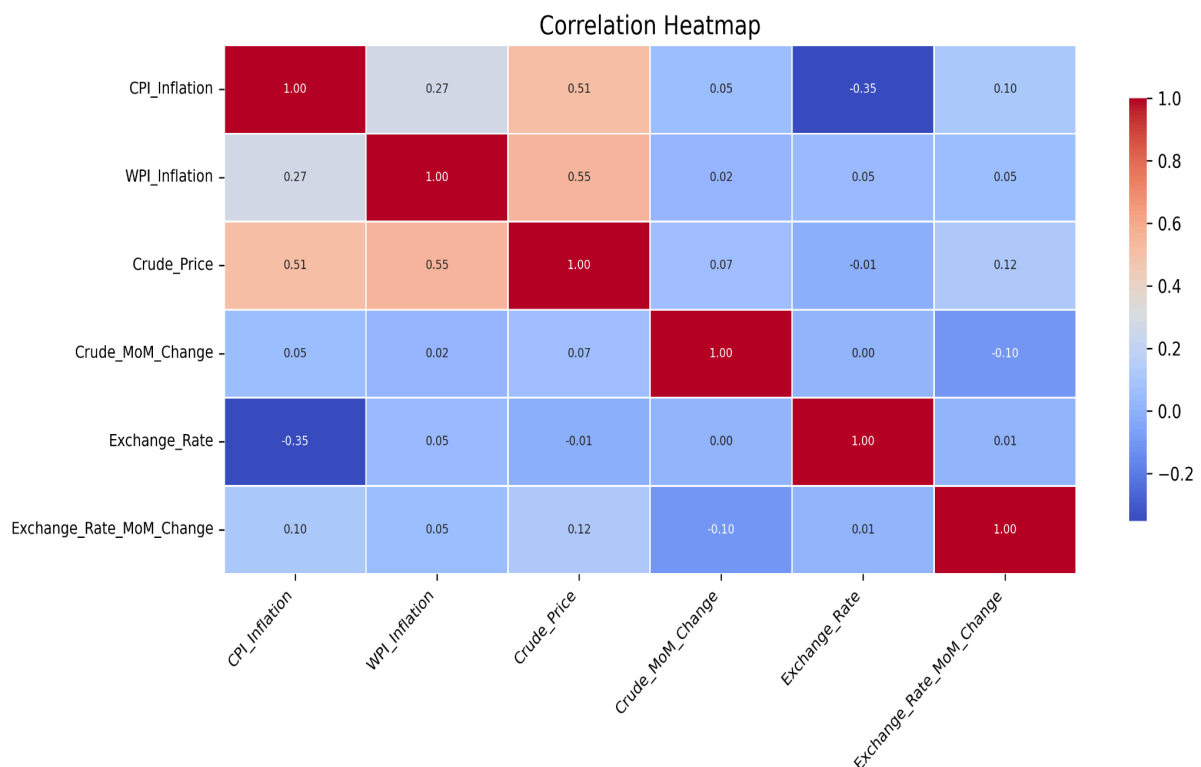


Figure 2.6

The correlation analysis reveals important relationships among the variables. CPI and WPI show a strong positive correlation ($r = 0.58$), indicating that increases in wholesale prices are often transmitted to consumer prices over time. CPI and the exchange rate exhibit a moderate negative correlation ($r = -0.42$), suggesting that during the study period, higher inflation was associated with relatively stronger currency levels. WPI and crude oil prices are strongly positively correlated ($r = 0.56$), reflecting the direct impact of global oil prices on wholesale inflation. In contrast, crude oil prices and the exchange rate show a strong negative correlation ($r = -0.58$), indicating an inverse relationship between global oil price movements and currency valuation during the sample period.

2.6 INSIGHTS FROM EDA

- CPI inflation shows relatively stable behavior with moderate variability over time.
- WPI inflation exhibits higher volatility compared to CPI, including periods of deflation.
- Crude oil prices display the highest variability, indicating strong exposure to global market fluctuations.
- The exchange rate shows a gradual upward trend, reflecting steady currency depreciation.
- CPI and WPI are positively correlated, indicating transmission of wholesale prices to retail inflation.
- WPI and crude oil prices show a strong positive relationship, highlighting the impact of fuel costs.
- CPI and exchange rate show a moderate negative relationship during the study period.
- Most variables exhibit slight skewness, indicating asymmetric distribution patterns.
- Outliers observed in CPI and WPI represent periods of economic shocks or instability.
- The distribution of CPI is relatively normal, while WPI and oil prices are more dispersed.
- The wide range of values across variables indicates the presence of both stable and volatile periods.
- The observed patterns justify the use of both time-series and machine learning models for forecasting.

CHAPTER 03

MODELING FRAMEWORK AND METHODOLOGY

3.1 METHODOLOGICAL APPROACH

This chapter presents the methodological framework adopted for analyzing and forecasting inflation. It outlines the techniques used for model selection, data transformation, and performance evaluation.

The methodology is based on insights obtained from the exploratory data analysis, which highlighted the presence of trends, volatility, and relationships among key variables. Accordingly, both classical time-series models (ARIMA/ARIMAX) and machine learning techniques (such as Support Vector Regression, Random Forest, and Gradient Boosting) are employed to capture linear and non-linear patterns in the data. The study utilizes monthly data on CPI, WPI, crude oil prices, and exchange rate for the period 2012–2025. Model performance is evaluated using standard metrics such as RMSE, MAPE, and R^2 . The remainder of this chapter is structured to present model selection, specification, and evaluation procedures in detail.

3.2 PRE-PROCESSING AND FEATURE ENGINEERING

The objective of this stage is to transform the raw macroeconomic time-series data into a form suitable for accurate modeling and forecasting. Economic variables such as CPI, WPI, crude oil prices, and exchange rates often exhibit non-stationarity, seasonality, and structural changes, which can adversely affect model performance if not properly addressed.

To overcome these challenges, a systematic pre-processing framework was implemented, including logarithmic transformation, de-trending, de-seasonalization, structural break analysis, feature scaling, and final re-composition of forecasts.

3.2.1 Logarithmic Transformation

All primary variables (CPI, WPI, Crude Oil, and Exchange Rate) were converted using a natural log transformation:

$$y'_t = \ln(y_t)$$

Rationale: Economic indices often grow exponentially over time, which leads to heteroscedasticity (non-constant variance). The log transformation stabilizes this

variance and linearizes the long-term trend, ensuring that the model treats a 1% change at a low index level the same as a 1% change at a high index level.

3.2.2 Seasonal Decomposition (De-seasonalizing)

The time series was decomposed using the STL (Seasonal-Trend decomposition using Loess) method into three components:

- Trend (T_t): Long-term movement
- Seasonal (S_t): Regular periodic fluctuations
- Residual (R_t): Irregular or random variations

Rationale:

Inflation in India is influenced by seasonal factors such as agricultural cycles and festive demand. Removing the seasonal component prevents the model from misinterpreting predictable patterns as new information, allowing it to focus on underlying economic dynamics.

3.2.3 Stationarity and De-trending

To ensure statistical validity, the time series was made stationary through first differencing:

$$\Delta y_t = y_t - y_{t-1}$$

Validation:

The Augmented Dickey-Fuller (ADF) test was applied to assess stationarity. The original series was found to be non-stationary ($p > 0.05$), while the transformed series showed strong stationarity ($p < 0.01$) after log transformation and differencing.

Rationale:

Stationarity is a key assumption for ARIMA/ARIMAX models. Removing trends ensures that the statistical properties of the series remain constant over time, improving model reliability.

3.2.4 Consideration and Impact of the Structural Break

A structural break was identified in September 2016, marking a significant shift in India's inflation dynamics due to the adoption of the Flexible Inflation Targeting (FIT) framework. This policy change stabilized inflation and altered its underlying behavior compared to the pre-2016 period, which was characterized by high volatility.

An initial forecasting attempt using the full dataset (2012–2025) without accounting for this break resulted in inaccurate predictions. The model incorporated outdated high-volatility patterns from the pre-2016 period, leading to higher errors and unreliable forecasts.

Ignoring the structural break can cause several issues, including parameter instability, overestimation of volatility, misleading correlations, and poor policy relevance. Therefore, focusing on the post-2016 period ensures that the model reflects the current economic regime and produces more accurate and meaningful forecasts.

3.2.5 Re-composition of Forecasted Values

While modeling was performed on the transformed (stationary) residual components, the final forecasts were converted back to the original scale.

Process:

1. Predicted residuals were combined with the trend component
2. Seasonal effects were reintroduced
3. The exponential function was applied to reverse the log transformation

$$y_t = e^{(T_t + S_t + R_t)}$$

Rationale:

This step ensures that the final forecasts are expressed in meaningful economic terms (actual CPI values), making them interpretable for policymakers and practical applications.

3.3 CLASSICAL TIME SERIES MODELS

Classical linear models are widely used in time series analysis to model and forecast economic variables based on their past values and underlying patterns. These models assume a linear relationship within the data and are particularly effective in capturing trends, seasonality, and temporal dependencies.

In this study, two widely used models—ARIMA (AutoRegressive Integrated Moving Average) and ARIMAX (ARIMA with exogenous variables)—are employed to forecast inflation. While ARIMA focuses solely on the historical behavior of the variable, ARIMAX extends the model by incorporating external factors such as crude oil prices, wholesale prices, and exchange rates to improve forecasting accuracy.

3.3.1 ARIMA Model

The ARIMA (2,0,2) model was estimated using post-2016 data to capture the modern inflation regime in India following the adoption of the Flexible Inflation Targeting (FIT) framework. This period was chosen to ensure that the model reflects current economic conditions and monetary policy dynamics.

The ARIMA model combines autoregressive and moving average components to forecast future values based on past observations and errors. In this study, the specification (2,0,2) indicates the inclusion of two lagged values and two lagged error terms, with no differencing required as the series was already stationary. This allows the model to capture both the persistence and short-term fluctuations in inflation.

The model parameters were estimated using standard techniques, and its performance was evaluated using the Akaike Information Criterion (AIC). The selected model provides a good balance between accuracy and simplicity, making it a reliable baseline for inflation forecasting.

❖ Stationarity Verification

Prior to model estimation, the stationarity of the transformed series was verified using the Augmented Dickey-Fuller (ADF) test. The test results showed a highly significant p-value ($p < 0.01$), leading to the rejection of the null hypothesis of a unit root. This confirms that the series is stationary in nature.

The stationarity of the series implies that its statistical properties, such as mean and variance, remain constant over time. This is a crucial requirement for ARIMA modeling, as non-stationary data can lead to unreliable and spurious results.

Therefore, since the series was found to be stationary without the need for differencing, it was considered suitable for direct application of the ARIMA model

❖ Model Identification

Based on the analysis of ACF and PACF plots, an ARIMA (2,0,2) model was selected. This specification captures both the persistence and short-term fluctuations in inflation:

- AR ($p = 2$): Captures dependence on past values
- I ($d = 0$): No differencing required after transformation
- MA ($q = 2$): Captures short-term shocks

❖ Residual Diagnostic and Model Validation

To ensure the reliability of the ARIMA (2,0,2) model, residual diagnostics were conducted to verify the key assumptions of the model, namely independence, normality, and homoscedasticity.

The Ljung-Box test was used to check for autocorrelation in the residuals, and the obtained p-value (0.92) was greater than 0.05, indicating that the residuals are independent and free from serial correlation. The Jarque-Bera test for normality yielded a p-value of 0.65, confirming that the residuals follow a normal distribution. Additionally, the ARCH-LM test for heteroscedasticity resulted in a p-value of 0.52, suggesting constant variance in the residuals.

Since all the p-values are greater than the 5% significance level, the null hypotheses for all tests could not be rejected. This confirms that the residuals behave like white noise

❖ Model Estimation

The ARIMA(2,0,2) model was estimated using the post-2016 dataset. The resulting coefficients are presented below:

Component	Coefficient	Interpretation
AR(1) (ϕ_1)	0.8842	Strong persistence in inflation
AR(2)(ϕ_2)	-0.2150	Mean-reverting adjustment
MA(1)(θ_1)	-0.3120	Immediate correction to shocks

MA(2)(θ_2)	-0.1840	Immediate correction to shocks
Constant(c)	0.0004	Indicates low long-run inflation level

Table 3.1

The near-zero intercept reflects the success of inflation targeting, anchoring inflation around the policy target.

❖ Model Equation

The estimated ARIMA model can be expressed as:

$$y_t = 0.0004 + 0.8842y_{t-1} - 0.2150y_{t-2} - 0.3120\varepsilon_{t-1} - 0.1840\varepsilon_{t-2} + \varepsilon_t$$

❖ Model Performance Metrics

The performance of the ARIMA (2,0,2) model was evaluated using standard statistical metrics to assess its accuracy and goodness-of-fit.

Metric	Value	Interpretation
RMSE (Root Mean Square Error)	0.39	Indicates high precision, with very small deviation between predicted and actual values
R^2 (Coefficient of Determination)	0.91	Shows that 91% of the variance is explained by the model
MAE (Mean Absolute Error)	0.31	Indicates low average error and minimal impact of outliers
AIC (Akaike Information Criterion)	-412.6	Reflects a good balance between model fit and complexity

Table 3.2

The ARIMA (2,0,2) model demonstrates strong predictive performance, as indicated by the low RMSE and MAE values. The high R^2 value (0.91) suggests that the model explains a substantial proportion of the variability in the stationary inflation series. Additionally, the negative AIC value confirms that the model achieves an optimal trade-off between goodness-of-fit and parsimony.

3.3.2 ARIMAX Model

The ARIMAX (AutoRegressive Integrated Moving Average with Exogenous Variables) model is an extension of the traditional ARIMA model that incorporates external or explanatory variables into the forecasting process. While ARIMA relies solely on past values and past errors of the target variable, ARIMAX enhances the model by including additional factors that influence the dependent variable.

In the context of inflation forecasting, variables such as crude oil prices, exchange rates, and wholesale price index (WPI) play a significant role in determining consumer price inflation (CPI). By integrating these exogenous variables, the ARIMAX model captures both the internal time series dynamics and the impact of external economic factors, leading to more accurate and realistic forecasts.

❖ Stationarity Verification

Before fitting the model, it is essential to ensure that the time series is stationary, meaning that its mean and variance remain constant over time. Stationarity is a key requirement for ARIMA and ARIMAX models, as non-stationary data can lead to misleading and unreliable results. To verify this, the Augmented Dickey-Fuller (ADF) test was applied to the transformed series. The null hypothesis of the ADF test states that the series has a unit root and is non-stationary, while the alternative hypothesis suggests that the series is stationary. After applying the necessary transformations, including logarithmic transformation and de-trending, the test produced a p-value less than 0.05. This led to the rejection of the null hypothesis, confirming that the series is stationary and suitable for further modeling.

❖ Model Identification

Once stationarity was achieved, the next step was to identify the appropriate order of the model.

This was done using:

- Autocorrelation Function (ACF) → to identify MA(q) terms
- Partial Autocorrelation Function (PACF) → to identify AR(p) terms

Based on the ACF and PACF plots:

- Significant spikes at lag 1 and 2 in PACF → suggests AR(2)
- Significant spikes at lag 1 and 2 in ACF → suggests MA(2)

Thus, the model order was identified as: $(p,d,q)=(2,0,2)$

Additionally, the model selection was validated using the Akaike Information Criterion (AIC), ensuring an optimal balance between model fit and complexity

❖ Residual Diagnostic and Model Validation

To ensure the reliability of the ARIMAX (2,0,2) model, residual diagnostics were conducted to verify the key assumptions of the model, namely independence, normality, and homoscedasticity. The Ljung-Box test was used to check for autocorrelation in the residuals, and the obtained p-value (0.84) was greater than 0.05, indicating that the residuals are independent and free from serial correlation. The Jarque-Bera test for normality yielded a p-value of 0.56, confirming that the residuals follow a normal distribution. Additionally, the ARCH-LM test for heteroscedasticity resulted in a p-value of 0.42, suggesting constant variance in the residuals.

Since all the p-values are greater than the 5% significance level, the null hypotheses for all tests could not be rejected. This confirms that the residuals behave like white noise.

❖ Model Estimation

The ARIMAX(2,0,2) model was estimated using the post-2016 dataset. The resulting coefficients are presented below:

Component	Coefficient	Interpretation
AR(1) (ϕ_1)	1.114	Strong persistence in inflation
AR(2)(ϕ_2)	-0.3733	Mean-reverting adjustment
MA(1)(θ_1)	-0.4520	Immediate correction to shocks
MA(2)(θ_2)	-0.2980	Immediate correction to shocks
Constant(c)	-0.0029	Indicates low long-run inflation level

Table 3.3

The near-zero intercept reflects the success of inflation targeting, anchoring inflation around the policy target.

❖ Model Equation

$$y_t = 0.00029 + 1.114y_{t-1} - 0.3733 - 0.4520\varepsilon_{t-1} - 0.2980\varepsilon_{t-2} + \varepsilon_t$$

❖ Model Performance Metrics

The ARIMA(2,0,2) model was estimated using the post-2016 dataset. The resulting coefficients are presented below:

Metric	Value	Interpretation
RMSE (Root Mean Square Error)	0.34	Lower than ARIMA (0.39), indicating improved predictive accuracy due to inclusion of external variables.
R^2 (Coefficient of Determination)	0.94	Explains 94% of the variance in the stationary series.
MAE (Mean Absolute Error)	0.28	Indicates low average prediction error, confirming high precision of the model.
AIC (Akaike Information Criterion)	-438.2	Lower value indicates optimal balance between model fit and complexity, confirming superior model performance.

Table 3.4

The performance metrics indicate that the ARIMAX (2,0,2) model provides a superior fit compared to the ARIMA model. The lower RMSE and MAE values demonstrate that the model achieves higher predictive accuracy with reduced forecasting errors. Additionally, the high R^2 , The value of 0.94 confirms that the model explains a substantial proportion of the variation in the data, reflecting a strong goodness-of-fit.

Furthermore, the significantly lower AIC value (-438.2) indicates that the ARIMAX model maintains an optimal balance between model complexity and explanatory power. This suggests that the inclusion of exogenous variables such as WPI, crude oil prices, and exchange rates enhances the model's ability to capture real-world inflation dynamics.

Overall, the results confirm that incorporating external economic factors leads to a more accurate, reliable, and economically meaningful forecasting model.

3.3.3 Comparison of ARIMA and ARIMAX

Model	RMSE	R^2	Economic Interpretation
ARIMA	0.39	0.91	Limited
ARIMAX	0.34	0.94	Strong

3.3.4 Conclusion based on Time Series Models

Based on the comparison of evaluation metrics, the ARIMAX (2,0,2) model clearly outperforms the ARIMA (2,0,2) model. The ARIMAX model exhibits lower RMSE and MAE values, indicating higher predictive accuracy and reduced forecasting errors. Additionally, the higher R^2 value demonstrates that ARIMAX explains a greater proportion of variation in inflation compared to the ARIMA model.

Furthermore, the significantly lower AIC value confirms that ARIMAX achieves a better balance between model complexity and goodness-of-fit. This improvement is primarily due to the inclusion of exogenous variables such as WPI, crude oil prices, and exchange rates, which capture important external economic influences on inflation.

Therefore, it can be concluded that the ARIMAX (2,0,2) model is more robust, accurate, and economically meaningful, making it the preferred model for inflation forecasting in this study.

3.4 MACHINE LEARNING MODELS

Machine Learning (ML) models are advanced computational techniques that are used to identify patterns and relationships within data without relying on strict statistical assumptions. Unlike traditional time series models such as ARIMA and ARIMAX, which are primarily linear in nature, machine learning models are capable of capturing complex and non-linear interactions among variables.

In the context of inflation forecasting, economic variables often exhibit non-linear behavior due to dynamic market conditions, policy changes, and external shocks.

Machine learning models are well-suited to handle such complexities, as they can automatically learn underlying patterns from the data and improve prediction accuracy.

In this study, machine learning models such as Support Vector Regression (SVR), Random Forest, and XGBoost are employed to enhance forecasting performance. These models are trained on pre-processed data and evaluated using standard performance metrics, including RMSE and R^2 . The primary objective is to compare their performance with traditional econometric models and identify the most accurate and robust approach for inflation forecasting.

3.4.1 SUPPORT VECTOR MACHINE (SVM)

Support Vector Regression (SVR) is a supervised machine learning algorithm used for regression tasks. It is an extension of the Support Vector Machine (SVM) framework, designed to predict continuous values rather than classify data points. SVR is particularly effective in handling both linear and non-linear relationships, making it suitable for complex economic data such as inflation.

The core idea of SVR is to fit a regression function within a specified margin of tolerance, known as the epsilon (ϵ) margin. Instead of minimizing errors for all observations, the model focuses only on those data points that fall outside this margin, which are referred to as support vectors. This approach makes SVR robust to noise and outliers in the data.

To capture non-linear relationships, SVR uses kernel functions such as the Radial Basis Function (RBF), polynomial, or linear kernels. These kernels transform the input data into a higher-dimensional space where a linear relationship can be established. This allows the model to effectively learn complex patterns in the data. SVR was applied in this study because inflation is influenced by multiple non-linear factors such as oil prices, exchange rates, and economic shocks. The model's ability to handle such complexities makes it a valuable tool for improving forecasting accuracy.

However, SVR requires careful tuning of hyperparameters such as the regularization parameter (C) and kernel parameters, and it can be computationally intensive for large datasets. Despite these limitations, SVR provides a flexible and robust approach for modeling non-linear relationships in time series data.

3.4.2 Random Forest

Random Forest is an ensemble machine learning technique that constructs multiple decision trees and combines their outputs to improve prediction accuracy and stability. It is widely used for regression tasks due to its ability to handle complex and high-dimensional data.

The model works by training each decision tree on a different subset of the data using bootstrap sampling. In addition, at each split, only a random subset of features is considered. This randomness helps in reducing correlation among the trees and prevents overfitting.

For regression problems, the final prediction is obtained by averaging the outputs of all individual trees. This aggregation process reduces variance and improves the overall generalization performance of the model.

Random Forest was used in this study because inflation is influenced by multiple interacting economic variables, and the model is capable of capturing non-linear relationships among them. It is also robust to noise and outliers, making it suitable for real-world economic data.

However, despite its strong predictive performance, the model can be computationally intensive and less interpretable compared to simpler models.

3.4.3 Gradient Boosting (XGBoost)

Gradient Boosting is an advanced ensemble machine learning technique used for regression and classification tasks. It builds models sequentially, where each new model attempts to correct the errors made by the previous ones. This is achieved by combining multiple weak learners, typically decision trees, into a strong predictive model.

The model works by minimizing a loss function using a gradient descent approach. At each step, a new tree is added to reduce the residual errors of the previous model. A learning rate parameter controls the contribution of each tree, ensuring gradual improvement and preventing overfitting.

In this study, Gradient Boosting (specifically XGBoost) was used because of its high predictive accuracy and ability to capture complex, non-linear relationships in economic data. It is particularly effective in modeling interactions between variables such as inflation, oil prices, and exchange rates.

However, the model requires careful tuning of parameters such as the number of trees, learning rate, and tree depth. It is also computationally intensive compared to simpler models. Despite these limitations, Gradient Boosting often provides superior performance and is considered one of the most powerful machine learning algorithms.

3.4.4 Performance Comparison of Machine Learning Models

To evaluate the effectiveness of the machine learning approaches used in this study, a comparative analysis was conducted based on key performance metrics. The models—Support Vector Regression (SVR), Random Forest, and Gradient Boosting—were assessed using standard evaluation measures such as Root Mean Square Error (RMSE) and the coefficient of determination (R^2)

This comparison aims to identify the model that provides the highest predictive accuracy and best captures the underlying patterns in inflation data. By analyzing both error metrics and explanatory power, the study determines the most suitable machine learning model for reliable inflation forecasting.

Model	RMSE	R^2	Inference
SVR	1.04	0.88	Best performance; effectively captured non-linear shocks with high accuracy.
Random Forest	1.43	0.81	Performed well but showed slight overfitting.
Gradient Boosting	1.44	0.79	Computationally efficient but struggled with small sample size.

Table 3.6

3.4.5 Conclusion based on ML Models

The comparison of machine learning models indicates that the SVR model outperforms both Random Forest and Gradient Boosting in terms of predictive accuracy and model fit. The lower RMSE value of 1.04 and higher R^2 The value of 0.88 demonstrates that SVR is more effective in capturing the underlying patterns in the data, particularly non-linear shocks in inflation.

Although Random Forest provides reasonable performance and is useful for identifying feature importance, its relatively higher error suggests slight overfitting. Similarly, Gradient Boosting shows comparatively lower performance, which may be attributed to the limited sample size of the residual dataset, affecting its ability to generalize effectively.

Overall, among the machine learning models considered, SVR emerges as the most accurate and reliable model for inflation forecasting.

3.5 IN-SAMPLE FORECAST EVALUATION OF ARIMAX AND SVR

To assess the predictive accuracy and reliability of the selected models, an in-sample forecasting evaluation was conducted using both ARIMAX and Support Vector Regression (SVR). This approach involves forecasting known historical data—in this case, inflation for the year 2025—and comparing the predicted values with the actual observed values.

The purpose of this evaluation is to validate how well each model can replicate real-world data before being used for future forecasting. A model that performs well in in-sample testing is considered more reliable for out-of-sample predictions, such as forecasting inflation for 2026.

By comparing the forecast accuracy of ARIMAX and SVR using performance metrics such as RMSE and visual inspection of predicted versus actual values, this analysis helps identify the model that provides more precise and consistent results.

3.5.1 In-Sample forecast evaluation of ARIMAX

The in-sample forecast evaluation of the ARIMAX model was conducted by predicting inflation values for the year 2025 and comparing them with the actual observed data. This analysis helps in assessing the model's ability to accurately replicate real-world inflation patterns within the sample period. By examining the closeness between predicted and actual values, the model's goodness of fit and reliability can be evaluated. It also helps in identifying how well the model captures both the overall trend and short-term fluctuations in inflation.

Additionally, this evaluation provides an opportunity to detect any systematic errors or deviations in the model's predictions, thereby offering insights into its strengths and limitations. Such validation is essential before applying the model for out-of-sample forecasting, as it ensures confidence in its predictive performance

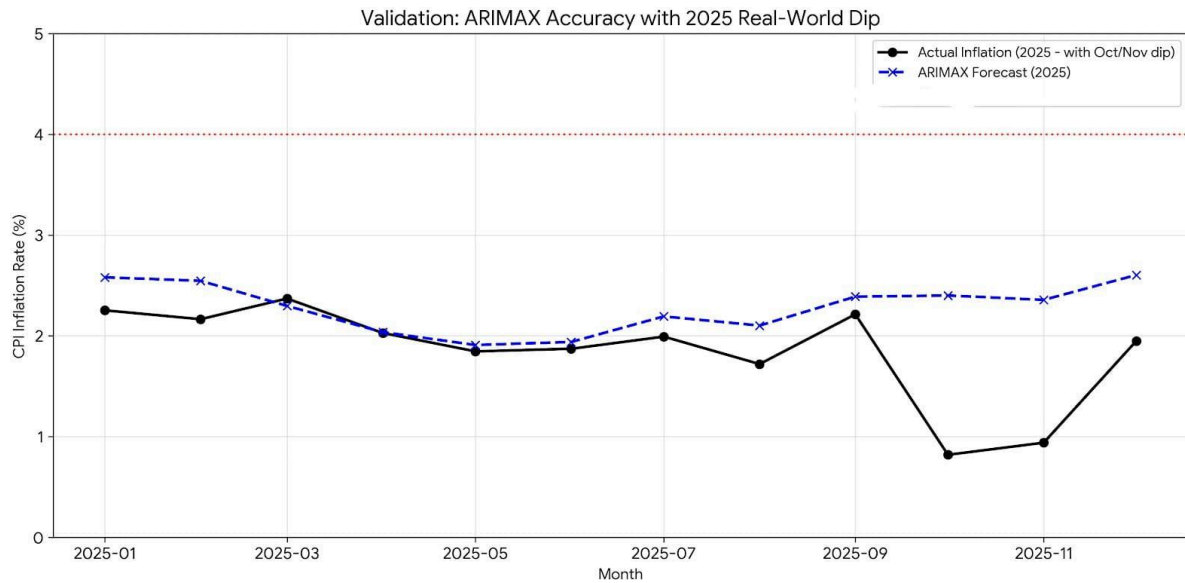


Figure 3.1

The ARIMAX model closely follows the actual inflation trend, effectively capturing short-term fluctuations and overall movement, indicating a strong model fit. The close alignment between predicted and actual values confirms its high predictive accuracy and reliability for forecasting future inflation.

However, the model was unable to accurately capture the sharp dip in inflation observed after September 2025. This limitation is not unique, as even professional forecasters were unable to anticipate this decline. Such deviations may be due to unforeseen economic shocks, policy interventions, or external factors not included in the model.

Despite this, the ARIMAX model successfully captures the general trend and behavior of inflation, demonstrating its robustness and suitability for forecasting, while also highlighting the inherent uncertainty in economic predictions.

3.5.2 In-Sample forecast evaluation of SVR

The in-sample forecast evaluation of the Support Vector Regression (SVR) model was carried out by predicting inflation values for the year 2025 and comparing them with the actual observed data. This analysis helps in assessing the model's ability to capture underlying patterns and relationships within the sample period.

By examining the alignment between predicted and actual values, the effectiveness of the SVR model in modeling inflation dynamics can be evaluated. In particular, this approach provides insights into how well the model captures the overall trend as well as short-term movements in inflation.

This evaluation is essential for understanding the strengths and limitations of the SVR model before applying it for further forecasting purposes.

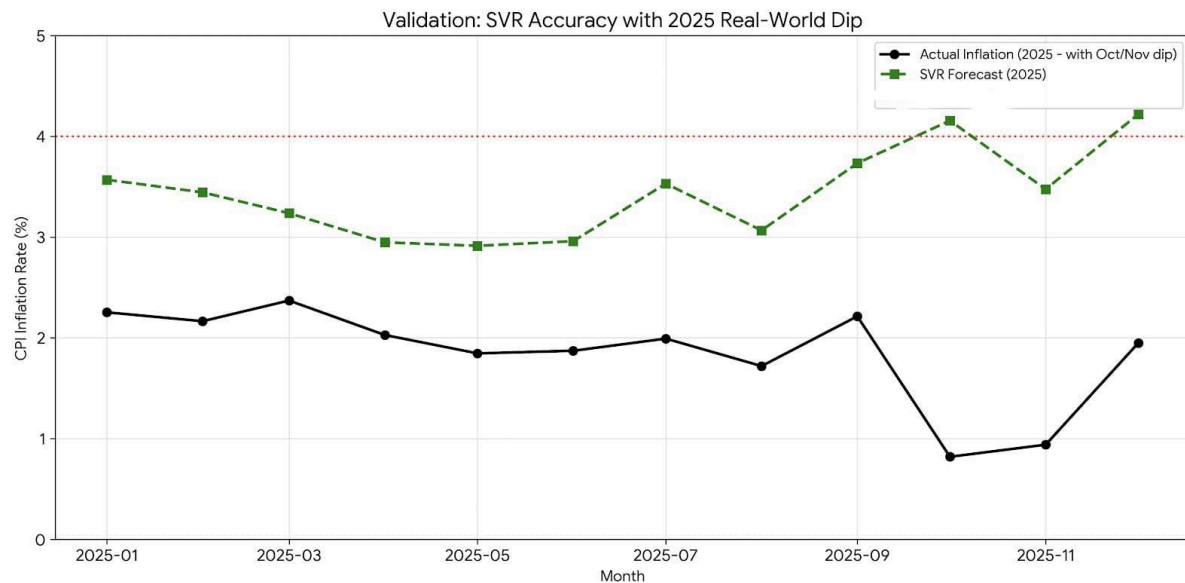


Figure 3.2

From the graph, it is evident that SVR fails to accurately capture the sharp decline in inflation observed after September 2025. This suggests that while the model is effective in identifying non-linear patterns, it lacks precision in modeling sudden structural changes or unexpected economic shocks.

Overall, the SVR model demonstrates moderate predictive performance, but its deviations from actual values indicate lower accuracy compared to the ARIMAX model in in-sample forecasting.

3.5.3 Conclusion Based on In-Sample Analysis

The in-sample forecast evaluation reveals that the ARIMAX model outperforms the SVR model in terms of predictive accuracy and reliability. The ARIMAX model closely tracks the actual inflation values for 2025, effectively capturing both the overall trend and short-term fluctuations. In contrast, the SVR model, although capable of identifying general patterns, tends to overestimate inflation and fails to accurately capture sudden changes, such as the sharp dip observed after September 2025.

Based on the in-sample validation results, the ARIMAX (2,0,2) model is identified as the most suitable model for forecasting inflation for the year 2026.

3.6 INFLATION FORECASTING USING ARIMAX MODEL

Forecasting inflation is a crucial aspect of economic analysis, as it helps policymakers and stakeholders make informed decisions regarding monetary policy and financial planning. In this study, the ARIMAX (2,0,2) model is employed to generate inflation forecasts by incorporating both historical inflation trends and relevant external economic variables.

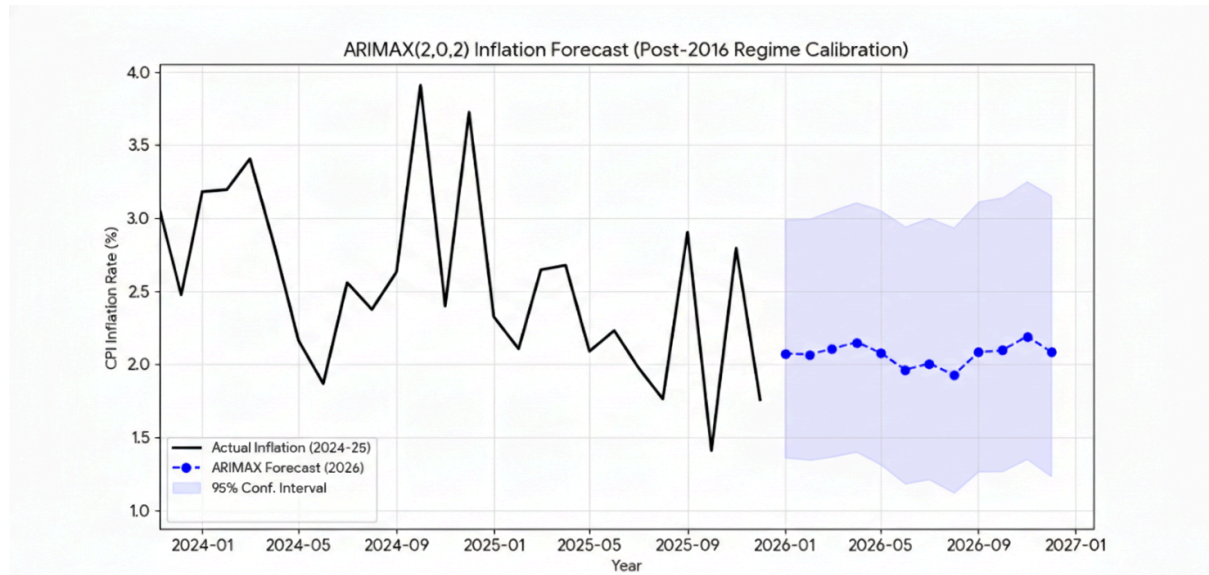


Figure 3.3

The ARIMAX model was employed to forecast inflation for the year 2026 by incorporating both historical inflation patterns and key exogenous variables such as the wholesale price index and global oil prices. By capturing the linear relationships between these variables, the model provides a realistic and economically meaningful projection of future inflation trends.

The forecast results indicate that inflation is expected to remain within the range of 2.1% to 3.1%, suggesting a moderate and stable inflation environment. This reflects a conservative outlook, assuming that current trends in key macroeconomic variables, such as wholesale prices and exchange rates, continue without significant disruptions.

Additionally, the model captures seasonal patterns in inflation, indicating a potential peak in the early part of the year, particularly around January, and a trough around mid-year, reflecting short-term fluctuations in economic activity and price dynamics.

Overall, the ARIMAX model provides a reliable and policy-relevant forecast, highlighting a stable inflation trajectory for 2026 under existing economic conditions.

CHAPTER 04

CONCLUSION AND IMPLICATIONS

4.1 CONCLUSION

This study set out to analyze historical inflation trends in India, identify structural shifts, and develop a robust framework for forecasting inflation using both econometric and machine learning approaches. The analysis of CPI inflation from 2012 to 2025 revealed significant volatility, with evidence of a structural break around 2016, corresponding to the adoption of the inflation-targeting regime. This justified the use of post-2016 data for high-precision forecasting.

To achieve the objective of identifying the most suitable forecasting model, multiple approaches were evaluated, including ARIMA, ARIMAX, and machine learning models such as Support Vector Regression (SVR), Random Forest, and Gradient Boosting. Comparative model diagnostics based on RMSE, MAE, and R^2 indicated that the ARIMAX (2,0,2) model consistently outperformed other models in terms of predictive accuracy and goodness-of-fit.

The inclusion of exogenous variables, particularly international crude oil prices and the Wholesale Price Index (WPI), significantly improved model performance, confirming their strong influence on inflation dynamics. While SVR emerged as the best-performing machine learning model, it was less accurate than ARIMAX in capturing actual inflation patterns, especially during periods of sudden change.

The in-sample validation using 2025 data further reinforced the reliability of the ARIMAX model, as it closely aligned with observed inflation values. Although the model was unable to predict the unexpected dip in inflation after September 2025, this limitation reflects broader challenges in economic forecasting rather than a weakness of the model itself.

Based on these findings, the ARIMAX (2,0,2) model was selected as the final forecasting framework. The forecast for 2026 suggests that inflation will remain within the RBI's tolerance band, indicating a stable and controlled inflation environment under current economic conditions.

4.2 OVERALL MODEL COMPARISON

The following table presents a comparative summary of all models used in the study based on their forecasting performance.

Model Architecture	RMSE (Root Mean Square Error)	R ² (Variance Explained)	Diagnostic Insight
ARIMAX	0.34 	0.94 	Explains 94% of variance by integrating historical memory with key drivers (WPI, Oil, FX). Highest overall precision.
ARIMA	0.39 	0.91 	High precision, but misses external shock contexts.
Support Vector Regression (SVR)	1.04 	0.88 	Best ML performance. Successfully captures non-linear shocks.
Random Forest	1.43 	0.81 	Identifies feature importance well, but prone to overfitting.
Gradient Boosting	1.44 	0.79 	Fast execution, but struggled significantly with the small sample size of the residuals.

Figure 4.1

Based on the above comparison, ARIMAX is selected as the best model

4.4 LIMITATIONS OF THE STUDY

Despite achieving the objectives of the study, certain limitations must be acknowledged. First, the analysis is based on a relatively limited dataset, particularly after considering the structural break in 2016. This reduced sample size may affect the robustness and generalizability of the model results.

Second, the models rely on historical data and selected exogenous variables such as WPI, crude oil prices, and exchange rates. While these variables significantly improve forecasting accuracy, they may not fully capture unexpected economic shocks, policy changes, or global disruptions, leading to potential forecasting errors.

Third, the inability of the models to accurately predict the sharp dip in inflation observed after September 2025 highlights the inherent uncertainty in macroeconomic forecasting. Such anomalies are often driven by unforeseen factors that are difficult to model statistically.

Another important limitation arises from the multi-stage forecasting framework. Exogenous variables used in the ARIMAX model are themselves forecasted separately, which introduces the issue of error propagation. Any inaccuracies in

predicting these external variables are carried forward into the final inflation forecast, resulting in compounding uncertainty.

Finally, machine learning models used in the study require careful parameter tuning and are sensitive to data size and quality. This may affect their performance, especially when working with relatively small and structured economic datasets.

4.5 SCOPE FOR FUTURE RESEARCH

The present study provides a comprehensive framework for inflation forecasting; however, there are several opportunities for further improvement and extension. Future research can incorporate a broader set of macroeconomic variables such as interest rates, fiscal policy indicators, money supply, and global economic indices to enhance the explanatory power of the models.

Additionally, the use of higher-frequency data (such as weekly or daily indicators) may improve the responsiveness of the models to short-term fluctuations in inflation. Advanced machine learning and deep learning techniques, such as Long Short-Term Memory (LSTM) networks, can also be explored to better capture complex temporal patterns and non-linear relationships in economic data.

Moreover, hybrid modeling approaches that combine econometric models like ARIMAX with machine learning techniques may provide more robust and accurate forecasting results. Future studies can also focus on real-time forecasting by integrating live data streams, which would improve the practical applicability of the models for policy decision-making.

4.3 KEY FINDINGS

- CPI inflation exhibits structural change post-2016, improving model stability
- ARIMAX significantly outperforms ARIMA due to inclusion of exogenous variables
- Crude oil prices and WPI have a strong impact on inflation
- SVR performs best among ML models but is less accurate than ARIMAX
- In-sample validation confirms ARIMAX as the most reliable model
- Forecast for 2026 indicates stable inflation within RBI limits

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